Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Justification

Annotation provenance is poorly documented across the benchmark. Only three datasets have explicit annotation documentation: DEFT-2021 (manual annotation, no annotator profiles) [Q25], CLISTER (multiple annotators, profiles undocumented) [Q41], and DiaMed (several annotators including one medical expert, but for ICD-10 classification only — not NER or STS) [Q45]. CAS POS was automatically annotated via Tagex 3 [Q42]. Author affiliations are academic NLP and clinical medicine institutions [Q8] — no regulatory-affairs or pharmacovigilance expertise is documented. The deployment explicitly anticipates systematic disagreement between biomedical NLP annotators and regulatory-standard judgments on borderline cases (A4). Empirical evidence in dataset analysis confirms this: high annotator variance on pharmaceutical pairs (DEFT2020-D10: scores [5,2,4,4,5]; DEFT2020-D11; DEFT2020-D22 [5,3,3,4,3]) shows non-specialist annotation; E3C systematically labels disease entities while leaving drug names unlabeled (E3C-D9, E3C-D14); QUAERO annotators were NLP/biomedical researchers mapping to UMLS, not regulatory affairs specialists (QUAERO-D20). OC is HIGH priority per the elicitation. The DiaMed inter-annotator agreement reporting [Q46] is a structural strength but does not address regulatory norms.

Strengths

- DiaMed reports inter-annotator agreement (Kappa Cohen, Gwet's AC1) computed across two annotation sessions [Q45, Q46]
- CLISTER averages multiple annotator scores producing graded ground truth [Q41], enabling variance analysis (DEFT2020-D10)
- CAS POS automatic annotations were validated against manual annotations at 98% precision [Q42]
- DiaMed includes one medical expert annotator [Q45] — partial domain expertise for the ICD-10 task

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels do NOT reflect regional regulatory stakeholder perspectives. No annotation guidelines reference EMA/ANSM regulatory standards. The deployment's authoritative ground truth rests with regulatory-affairs specialists, but no benchmark dataset documents such annotators (QUAERO-D20, CLISTER-D16).

ID	Evidence
QUAERO-D20	`"produit dans une lignée cellulaire murine par génie génétique ."` — Murine cell line tagged as LIVB; biologic/regulatory classification may differ from UMLS grouping

CLISTER-D16	Le sevrage sera obtenu lors d'une hospitalisation au Centre Antipoison de Marseille, avec baisse progressive des doses quotidiennes. / Un sevrage sera effectué après 12 jours d'hospitalisation au Centre Antipoison de Marseille. — Duration detail (progressive vs. 12 days) scored 4.0; regulatory annotator might score lower

OC-2: Systematic disagreement is empirically expected: annotator variance on pharmaceutical STS pairs (DEFT2020-D10 [5,2,4,4,5]; DEFT2020-D22 [5,3,3,4,3]) demonstrates non-specialist annotation; E3C consistently labels diseases while leaving drug names O (E3C-D9, E3C-D14) reflects clinical-priority norms divergent from regulatory; A4 from elicitation explicitly anticipates this gap.

ID	Evidence
DEFT2020-D10	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High annotator variance (range 2–5) on opioid timing safety statement
DEFT2020-D22	Chez les patients insuffisamment équilibrés par glimepiride arrow à la dose maximale... — Dosage instruction pair; 2-point annotator variance on pharmaceutical content
E3C-D9	"ciclosporine, d'evérolimus et de corticoïdes" unlabeled; "insuffisance rénale modérée" labeled CLINENTITY — Annotation prioritizes disease over drug entities
E3C-D14	ciclosporine, d'evérolimus et de corticoïdes ... insuffisance rénale ... anémie ... hypertension artérielle — Disease entities labeled, drug names not — clinical vs. regulatory annotator priority gap
WEB-12	Bayer et al. 2021 Drug Safety confirms general biomedical NLP systems 'would need to be modified or reconfigured to lower error rates to support their use in a regulatory setting'

OC-3: Annotator demographics are NOT DOCUMENTED for the majority of corpora. CLISTER says 'multiple annotators' [Q41] without profiles; DiaMed lists 'several annotators including one medical expert' [Q45]; QUAERO, E3C, PxCorpus, ESSAI annotator profiles are not documented in the paper. No Datasheets-style demographic disclosures exist.

OC-4: Re-annotation by regulatory-affairs annotators would be required for deployment-grade label quality. No such re-annotation effort is documented in DrBenchmark or in any published French NLP corpus (gap_id 4 RESOLVED as confirmed gap).

OC-5: Aggregation by averaging in CLISTER [Q41] erases minority/dissenting judgments — visible as variance pattern (DEFT2020-D10). For regulatory equivalence, dissenting annotators flagging legal mismatch would be the signal of interest, but the averaging method suppresses it.

ID	Evidence
DEFT2020-D10	Ce produit peut provoquer un syndrome de sevrage opiacé s'il est administré à un toxicomane moins de 4 heures après la dernière prise de stupéfiant. — High annotator variance (range 2–5) on opioid timing safety statement

OC-6: Label issues harming convergent and external validity: (a) annotator population systematically misaligned with deployment ground-truth source; (b) no EMA/ANSM annotation guidelines documented; (c) variance suppression via averaging in STS; (d) consistent clinical-priority annotation norms in NER (E3C-D9, QUAERO-D18); (e) no documented re-annotation effort exists for regulatory norms.

ID	Evidence
DEFT2020-D18	...contre-indiqué en cas de galactosémie congénitale..." vs. "...ne doit pas être utilisé en cas de galactosémie... — "Congénitale" qualifier dropped in target; scored 4.6 despite regulatory-material difference
QUAERO-D18	`"La dose quotidienne maximale est de 21 , 6 microgrammes par jour ."` — Specific dose value and unit are unnanotated (O tags); dosage not a labeled entity class

Information Gaps

- No published comparison of biomedical NLP annotator vs. regulatory affairs expert judgments on EMA-format text (gap_id 4 RESOLVED as confirmed gap)
- Annotator professional background documentation absent for QUAERO, E3C, ESSAI, PxCorpus, MORFITT, MANTRAGSC, DEFT2020, DEFT2021

Requires Expert Verification

- Magnitude of expected disagreement between benchmark labels and regulatory-affairs judgments on contraindication-qualifier and dose-threshold borderline cases — would require pilot re-annotation
- Whether DiaMed's medical expert participation translates to any regulatory-affairs annotation norm overlap

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Remediation

Gap 1: No annotation guidelines reference EMA/ANSM standards; annotator populations are academic NLP and clinical, with no documented regulatory-affairs expertise; STS averaging suppresses dissent that would signal regulatory mismatch.

Recommendation: Commission a pilot re-annotation of 200–500 regulatory-relevant items (QUAERO EMEA NER, CLISTER and DEFT2020 STS pairs containing dose/contraindication content) by regulatory-affairs specialists. Quantify the disagreement rate with benchmark labels on borderline cases. Use this to bound expected error in deployment and to calibrate the human-review flagging threshold.